BRIDGE-DN — Communication Protocol & Dongle Reference

A complete coverage map for the BRIDGE-DN hub: every smart-home communication technology worth supporting, the dongle that covers each one (flagship + budget solo-purpose alternative), mesh-node expansion options, and the Home Assistant software stack needed to run it all.

Design philosophy for this build: **solo-purpose dedicated dongles as the baseline** (one radio, one job — better isolation, easier troubleshooting, no MultiPAN instability), with an optional **multi-protocol flagship** noted separately where you'd prefer to consolidate.

Key shopping principle throughout: **keep all 2.4 GHz radios (Zigbee, Thread, BLE) on USB 2.0 with extension cables away from the host and from USB 3.0 ports.** USB 3.0 emits broadband 2.4 GHz noise that desensitizes these radios. This single rule prevents the majority of "my mesh is flaky" problems.

PART 1 — Protocol-by-Protocol Coverage

Each entry: what it is, the core specs to shop by, ideal hardware placement, a flagship dongle, and a cheaper solo-purpose alternative.

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1. Wi-Fi (2.4 / 5 GHz)

What it carries: The single largest category of consumer smart devices — Tapo (incl. your H110), Shelly Pro, most cameras, most smart plugs, and Matter-over-Wi-Fi devices. Also the transport your whole HA instance rides on.

Base specs to shop by:

- Dual-band (2.4 GHz *and* 5 GHz) — many cheap adapters are 2.4-only

- Modern standard: Wi-Fi 5 (802.11ac) minimum, Wi-Fi 6 (802.11ax) preferred for future-proofing
- Linux driver maturity — this is the real gotcha. Look for chipsets with mainline kernel drivers (MediaTek MT7612U, MT7921U, or Realtek RTL8812AU with known driver support)
- AP-mode capability if you ever want the host to broadcast its own commissioning network

Ideal placement: Onboard the mini PC / Pi is usually sufficient since Wi-Fi is IP-routed (not latency-critical like Z-Wave). If using a USB adapter, either hub is fine — Wi-Fi tolerates USB 3.0 better than the mesh radios, but a short extension away from the metal box still helps.

Flagship (multi-purpose): *Onboard Pi 5 / mini-PC Wi-Fi+BT* — for most builds you don't buy a Wi-Fi dongle at all; the host's built-in radio covers it. If you need external: **Alfa AWUS036AXML** (Wi-Fi 6, dual-band, MT7921U, strong Linux support).

Budget solo-purpose: TP-Link Archer T3U / T2U Nano (dual-band AC, small, cheap) or an **MT7612U-based generic dongle** (~€12, reliable Linux driver).

Mesh? Wi-Fi itself is star-topology (not device mesh in the Zigbee sense), though Wi-Fi *mesh router systems* exist at the network layer. Not a mesh you extend with cheap nodes for smart-home purposes.

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2. Bluetooth LE (Bluetooth Low Energy)

What it carries: BLE sensors (temperature/humidity like Xiaomi LYWSD03MMC, SwitchBot devices, many plant sensors), **and — critically — the commissioning channel for almost every Matter device.** Matter devices are first paired over BLE before handing off to Wi-Fi or Thread. Without BLE, Matter onboarding is painful.

Base specs to shop by:

- Bluetooth 5.0+ (5.2 / 5.3 / 5.4 preferred for LE range and throughput)
- BLE (Low Energy) explicitly supported, not just classic Bluetooth
- Chipset with Linux BlueZ support — Realtek RTL8761B (BT 5.x) is the safe default
- Antenna: external/whip antenna versions reach noticeably further than nano/flush versions

Ideal placement: USB 2.0, extension cable, positioned centrally if you rely on it for BLE sensors. For commissioning-only use, onboard BT is fine.

Flagship (multi-purpose): covered by the onboard Pi 5 radio or an *ESP32* acting as a distributed **Bluetooth Proxy** (see Part 2 — this is the better pattern for whole-home BLE than one big dongle).

Budget solo-purpose: TP-Link UB500 (BT 5.4, nano, RTL8761B, ~€10, plug-and-play on Linux) or **UGREEN BT 5.3 adapter** (external antenna variant for range).

Mesh? Bluetooth *Mesh* exists as a formal spec but is rare in consumer HA setups. BLE for HA purposes is effectively extended via **ESP32 Bluetooth Proxies**, not native mesh — cheap, and the recommended approach.

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3. Zigbee (802.15.4, 2.4 GHz mesh)

What it carries: The workhorse. Philips Hue, IKEA Trådfri (bulbs/remotes), Aqara, SONOFF, frient, Third Reality, Develco — thousands of devices. Your DIN relay modules (frient, SONOFF DIN, Aqara T2) live here too.

Base specs to shop by:

- Chipset: **Silicon Labs EFR32MG24** (current best) or MG21 (still excellent); or Texas Instruments **CC2652P** (the SONOFF budget standard). Avoid older CC2531.
- **External antenna** strongly preferred over PCB-trace for coordinator role

- **USB 2.0 interface** (never buy a USB 3.0 Zigbee stick — the point is to stay off that noisy bus)
- Firmware: coordinator-capable, flashable, supported by ZHA and Zigbee2MQTT
- Comes with (or you add) a **USB extension cable** — non-negotiable for coordinators

Ideal placement: USB 2.0 port, on a 1–2 m extension cable, out of the metal distribution box if possible, elevated and clear of other electronics. Placement is the #1 determinant of Zigbee mesh health.

Flagship (multi-purpose — Zigbee OR Thread): Home Assistant Connect ZBT-2 (Nabu Casa). Silicon Labs MG24 + ESP32-S3 bridge, 4.16 dBi tuned external antenna, 460,800 bps, ~€49. One caveat central to your solo-purpose philosophy: *it does Zigbee **or** Thread, never both at once* — Nabu Casa explicitly dropped MultiPAN as unreliable. So even the flagship is effectively solo-purpose per unit; buy two if you want both.

Budget solo-purpose: SONOFF Zigbee 3.0 USB Dongle Plus — the *-P* (CC2652P) or *-E* (EFR32MG21) variant, ~€20, the single most widely deployed HA Zigbee coordinator. Add a USB extension.

Mesh? Yes — core mesh technology. Mains-powered Zigbee devices (bulbs, plugs, DIN relays) automatically act as routers, extending the mesh. See Part 2 for cheap dedicated router nodes.

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4. Thread (802.15.4, 2.4 GHz mesh — the IP-native sibling of Zigbee)

What it carries: The transport for modern **Matter-over-Thread** devices — Eve, Nanoleaf, newer IKEA sensors (which are Thread-only, no Zigbee fallback), Aqara's newer line, Apple/Google Thread accessories. Same radio silicon as Zigbee, different network stack (IPv6/6LoWPAN).

Base specs to shop by:

- Same MG24 / MG21 chipset family as Zigbee
- Must be flashable with **OpenThread RCP firmware** and run as a **Thread Border Router**
- External antenna, USB 2.0, extension cable — identical hygiene to Zigbee
- HA support via the **OpenThread Border Router** add-on

Ideal placement: USB 2.0 + extension, same as Zigbee. If you run both Zigbee and Thread coordinators, **physically separate the two antennas** and, ideally, pin them to different 2.4 GHz channels to reduce mutual interference.

Flagship (multi-purpose): A **second Home Assistant Connect ZBT-2**, this one dedicated to Thread. (Same hardware as your Zigbee unit, flashed for the other protocol — clean, identical spares.)

Budget solo-purpose: SONOFF Zigbee 3.0 USB Dongle Plus MG24 flashed with Thread/OpenThread firmware (~€25), or an **nRF52840 dongle** (Nordic) flashed as an OpenThread RCP — the DIY route.

Mesh? Yes — core mesh technology, and uniquely supports *multiple border routers* on one Thread network for redundancy (unlike Zigbee's single-coordinator model). This is a genuine architectural advantage — see Part 2.

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5. Z-Wave (sub-GHz mesh — 868.42 MHz in EU)

What it carries: Aeotec, Fibaro, Qubino, Zooz, Heltun, Shelly Wave, Ultraloq — a mature, ~4,500-device ecosystem. Its sub-GHz frequency **penetrates walls better and avoids the congested 2.4 GHz band entirely** — valuable in a masonry Budapest building.

Base specs to shop by:

- **800-series chipset** (Silicon Labs ZG23) — most future-proof, supports Z-Wave Long Range. 700-series acceptable but not recommended new; avoid 500-series for

new builds.

- **Correct regional frequency: 868.42 MHz for EU.** A US 908 MHz stick will not talk to EU devices — this is the single most common ordering mistake.
- **S2 security + SmartStart** support
- USB connection (not a module — USB passes through virtualization/Docker far more cleanly)
- **Latency-sensitive: keep this one directly on the host via USB**, not over a network bridge, unless you run Z-Wave JS remotely (see notes)

Ideal placement: USB, upright, antenna clear of obstructions. Because Z-Wave is 868 MHz it does *not* conflict with your 2.4 GHz radios — it can sit nearer the host without the USB 3.0 worry, though a short extension still helps. Nabu Casa's flagship even has an accelerometer that blinks if you place it on its side.

Flagship (multi-purpose-ish): Home Assistant Connect ZWA-2 (Nabu Casa). Silicon Labs ZG23 800-series, tuned high-gain antenna, Z-Wave Long Range, S2/SmartStart, auto-region-detect (defaults to EU), ~€59–69. Single global SKU — no wrong-region risk.

Budget solo-purpose: Zooz 800 Series ZST39 (get the **EU 868 MHz** variant, *not* the ZST39 LR which is US-frequency), ~€30. Or **Aeotec Z-Stick 7 (EU, ZWA010)**.

Mesh? Yes — mesh technology. Mains-powered Z-Wave devices act as repeaters. 800-series adds Long Range (star-topology point-to-point for extreme distance). Cheap dedicated repeater nodes exist — see Part 2.

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6. Matter (application layer — rides over Wi-Fi, Thread, or Ethernet)

What it is: *Not a radio.* Matter is a unifying application/language layer that runs **over** an existing transport. This is the cross-option you specifically wanted covered. You do **not** buy a "Matter dongle."

The cross-options ("Matter-over-..."):

- **Matter-over-Wi-Fi** → covered by your **Wi-Fi** adapter (#1). This is how the H110 connects.
- **Matter-over-Thread** → covered by your **Thread** border router (#4).
- **Matter-over-Ethernet** → covered by the host's **Ethernet** NIC (some Matter devices are wired).
- **BLE commissioning** → covered by your **Bluetooth** adapter (#2), used to onboard all of the above.

Base specs to shop by: Nothing to buy — instead ensure your HA has the **Matter Server** add-on and that you have working Wi-Fi + Thread + BLE. That trio = full Matter coverage.

Ideal placement: N/A (software layer). Just ensure the underlying radios are well-placed.

Mesh? Inherits mesh from its transport: Matter-over-Thread is meshed; Matter-over-Wi-Fi is not.

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7. Infrared (IR) — legacy but ubiquitous

What it carries: Every TV, air conditioner, soundbar, projector, and set-top box with a remote. In your building's context: split-unit AC control is the killer app.

Base specs to shop by:

- Transmit power / number of IR LEDs and spread (for whole-room coverage)
- Learning capability (to capture unknown remote codes) vs database-only
- Local control (avoid cloud-only blasters); native HA or ESPHome support
- Combo IR+RF433 units cover two legacy bands at once

Ideal placement: Line-of-sight to target devices — so **not** in the distribution box. This is inherently a distributed, in-room device (network-connected, not a host dongle).

Flagship (multi-purpose IR+RF): BroadLink RM4 Pro (IR + 433 MHz RF, learning, large code database, HA integration).

Budget solo-purpose: LinknLink eRemote (~€13, native MQTT, local) or a **DIY ESP32 + IR LED running ESPHome** — cheapest, fully local, and plays to your ESP32 strengths. Note: **HA gained native IR support in the 2026.4 release**, following the Bluetooth-proxy model, so ESP32 IR proxies are now first-class.

Mesh? No. IR is point-to-point line-of-sight. "Coverage" = more blasters, not a mesh.

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8. Sub-GHz RF 433 / 868 / 315 MHz (legacy ISM-band OOK/ASK)

What it carries: The huge world of cheap legacy RF — old wireless doorbells, weather stations, garage remotes, budget RF plug sockets (Lidl/aldi-type), PIR alarm sensors, TPMS, some utility meters. Receive-mostly (monitoring), with transmit for controlling old RF sockets/blinds.

Base specs to shop by:

- **For receive/monitor:** an **RTL-SDR** (RTL2832U + R820T2 tuner) covering ~500 kHz–1.7 GHz, decoded in software by `rtl_433`
- **For transmit** (control old RF outlets, Somfy blinds): a transceiver like the **RFXtrx433E/XL** or a **CC1101-based** board
- Antenna tuned to the band you care about (433 MHz most common in EU consumer gear)

Ideal placement: USB. The SDR is not latency-critical and not 2.4 GHz, so it can live on the **USB 3.0 hub** happily — a rare exception to the "everything on 2.0" rule.

Flagship (multi-purpose TX+RX): RFXtrx433E (transmit + receive, supports LightwaveRF, Somfy RTS blinds, many sensor brands; fires HA events).

Budget solo-purpose: Nooelec NESDR SMArt / Mini 2 (~€35, receive/monitor via rtl_433) or a **generic RTL2832U dongle** (~€15).

Mesh? No — broadcast RF, no mesh. Coverage via antenna quality/placement, or multiple SDR receivers feeding one HA.

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9. Legacy Bluetooth Classic (2.x / 3.x)

What it carries: Old BT audio, some older BT peripherals and beacons predating BLE. Niche, but "anything with wireless on the box."

Base specs to shop by: Backward-compatible BT adapter (BT 4.0+ chipsets cover classic + LE); CSR8510 or RTL8761B chipsets.

Ideal placement: USB 2.0. Largely subsumed by any modern BLE adapter (#2) — a BT 5.x dongle covers classic too.

Flagship / budget: Same **TP-Link UB500** or **ASUS USB-BT500** covers both classic and LE — no separate purchase needed.

Mesh? No.

FUTURE-PROOFING — protocols to leave a port and budget for

These aren't fully shippable dongles yet (early/certifying in 2026), but the build should anticipate them.

10. Zigbee 4.0 / Sub-GHz "Suzi" (868/915 MHz long-range Zigbee)

Zigbee expanding beyond 2.4 GHz into sub-GHz for long range and wall penetration —

competing with Z-Wave LR and Wi-Fi HaLow. Certification opening H1 2026. **Requires new radio hardware** (can't firmware-update a 2.4 GHz stick into it). Action: leave a spare USB 2.0 port earmarked for a Suzi-capable coordinator once certified hardware ships. Also brings **Zigbee Direct** (BLE-assisted pairing).

11. Wi-Fi HaLow (802.11ah, sub-1 GHz)

Sub-1 GHz Wi-Fi: >1 km range, years of battery life, native IP. Sits between Zigbee/BLE and cellular IoT. Still niche; would need its own USB chipset distinct from your 2.4/5 GHz Wi-Fi adapter. Watch given your LoRa/Meshtastic interests. Action: monitor, no purchase yet.

12. LoRa / Meshtastic (sub-GHz long-range, ~868 MHz EU)

Not a mainstream HA smart-home protocol, but you already run it. Bridgeable into HA via MQTT. Action: a **LoRa/Meshtastic node (e.g. Heltec/LILYGO T-Beam)** can feed HA over MQTT — relevant for perimeter/garden/outbuilding telemetry where even Z-Wave LR won't reach.

PART 2 — Mesh Map & Cheap Node-Expansion Devices

Which technologies mesh, and the cheapest dedicated devices to drop in as extra nodes to thicken/extend each mesh layer.

Protocol	Mesh?	How it extends	Cheapest dedicated node to add a layer
Wi-Fi	Network-mesh only	AP/mesh routers	(not a device-mesh; use more APs)
Bluetooth LE	Via proxies	ESP32 Bluetooth Proxy pattern	**ESP32 dev board (~€5)** flashed with ESPHome Bluetooth Proxy — scatter one per room
Zigbee	✅ Native mesh	Any mains-powered router device repeats	**SONOFF ZBMINI / IKEA Trådfri plug / any cheap Zigbee smart plug (~€8–12)**; or **SONOFF ZBDongle flashed as router**
Thread	✅ Native mesh + multi-border-router	Any mains Thread device routes; add border routers for redundancy	**Any always-powered Matter-over-Thread plug**; extra **ZBT-2 / SONOFF MG24** as additional border routers
Z-Wave	✅ Native mesh	Any mains-powered Z-Wave device repeats	**Cheap Z-Wave mains plug (Aeotec/Zooz range, ~€25)** as a repeater; or Z-Wave LR for point-to-point reach
IR	❌	More blasters	**ESP32 + IR LED (ESPHome, ~€6)** per room
Sub-GHz 433	❌	More receivers	Extra **RTL-SDR (~€15)** feeding same HA

Protocol	Mesh?	How it extends	Cheapest dedicated node to add a layer
Matter	Inherits transport	Via Thread mesh / Wi-Fi	(see Thread / Wi-Fi rows)

The distributed-node insight for your build: rather than one dongle-heavy host trying to reach the whole apartment from the distribution box, the cheapest and most robust "next level" is **ESP32 nodes (~€5 each) running ESPHome** scattered per room — each acting simultaneously as a **Bluetooth Proxy** and, if wanted, **IR blaster** and sensor node, all reporting to HA over Wi-Fi. This solves the wall-penetration problem that dongles-at-the-box can't, cheaply, and plays directly to your ESPBLOK direction.

Placement summary for the two-hub split:

- **USB 2.0 hub (radios, extension cables, away from box):** Zigbee coordinator, Thread border router, BLE adapter
- **USB 3.0 hub (bandwidth/non-2.4 GHz):** Wi-Fi adapter (if external), RTL-SDR, storage, Ethernet adapter
- **Z-Wave:** directly on host USB (latency-sensitive, 868 MHz so no 2.4 GHz conflict)
- **IR / distributed BLE / sensors:** ESP32 proxies in-room, over Wi-Fi — not in the box at all

DIN form-factor note (secondary): Almost none of these radio coordinators exist as true screw-terminal DIN modules — the ecosystem is dongle/network-device based. The realistic DIN-adjacent option is the **SMLIGHT SLZB-06 / SLZB-06M / SLZB-MR** family (PoE/USB/LAN Zigbee+Thread coordinators) which are small enough to fix to a DIN backplate or a generic DIN bracket, and give you Ethernet/PoE placement freedom. Z-Wave, BLE, and SDR have **no** DIN-native form; they stay as USB dongles on the host. So: DIN modularity applies to the *coordinators-over-Ethernet* (SLZB family) but not to the direct-USB radios — consistent with the dongle-based approach being primary.

PART 3 — Home Assistant Software Stack (one by one)

Everything needed on the HA side to actually run the above. Grouped by role. In HA-OS these are "Add-ons"; in Docker/Container installs they're separate containers.

Core platform

1. **Home Assistant** (Core + OS, or Core in Docker) — the controller itself. HA-OS recommended for the DIN Pi build (bare-metal, headless, auto-boots into HA).
2. **Mosquitto (MQTT broker)** — the message bus that ties Zigbee2MQTT, ESPHome nodes, and many integrations together. Install this early; lots of things depend on it.

Zigbee (choose ONE controller)

3a. **Zigbee2MQTT** — the more powerful/flexible Zigbee stack; publishes to MQTT (needs Mosquitto). Broadest device support, best for a large mixed-brand mesh. **Recommended for your multi-brand scope.** 3b. **ZHA (Zigbee Home Assistant)** — native integration, no MQTT needed, simpler. Alternative to Z2M — don't run both on the same coordinator.

Thread + Matter

4. **OpenThread Border Router** (add-on) — runs your Thread mesh on the dedicated Thread dongle.
5. **Matter Server** (add-on) — the Matter controller/fabric; connects to HA over WebSocket. Required for **all** Matter devices (over Wi-Fi *and* over Thread), including the H110.

Z-Wave

6. **Z-Wave JS UI** (or the **Z-Wave JS** add-on) — the Z-Wave controller software. UI

variant gives a fuller management interface. Latency-sensitive link to the ZWA-2/Zooz stick — keep on USB.

Bluetooth

7. **Bluetooth integration** (built into HA core) — for the local BLE adapter.
8. **ESPHome** (add-on) — manages your ESP32 nodes, including **Bluetooth Proxy** firmware for whole-home BLE, plus IR blasters and sensors. Central to the distributed-node strategy.

Legacy RF & IR

9. **rtl_433** (via the **RTL_433 MQTT Auto Discovery** add-on) — decodes 433/868/315 MHz sensors from the RTL-SDR into MQTT/HA.
10. **RFXtrx integration** (built into HA core) — for the RFXtrx433E transceiver (TX+RX of old RF gear).
11. **Broadlink integration** (built into HA core) — for the RM4 Pro IR/RF blaster. (Or ESPHome IR, or HA's native 2026.4 IR support.)

Supporting / operational

12. **Node-RED** (add-on) — advanced visual automation/flow logic beyond HA's built-in automations (you already use this in the BRIDGE project).
13. **File editor / Studio Code Server** (add-on) — for editing YAML config on the headless box.
14. **Samba / SSH** (add-ons) — remote file access and shell to the DIN Pi.
15. **Backup** (built-in) — schedule these; a distribution-box Pi you don't want to re-provision by hand.

Optional depending on scope

16. **KNX integration** (built into HA core) — only if you add a KNX IP gateway for building-automation DIN modules.
17. **Modbus integration** (built into HA core) — for RS-485 energy meters (Eastron SDM630, etc.) via the RevPi's native RS-485 or an Elfin EW11 bridge.
18. **Mumble/Murmur, Meshtastic-to-MQTT** — bridges for your intercom and LoRa props, if folded into HA.

One-line dependency chain to remember

`Mosquitto (MQTT)` underpins → `Zigbee2MQTT`, `ESPHome nodes`, `rtl_433`. `Matter Server` needs → working `Wi-Fi` + `OpenThread Border Router` + `Bluetooth`. `Z-Wave JS UI` is standalone but latency-bound to a **USB** connection.

Install order that avoids backtracking: **HA-OS → Mosquitto → (Zigbee2MQTT / ZHA) → OpenThread BR → Matter Server → Z-Wave JS UI → ESPHome → rtl_433 / RFXtrx / Broadlink → Node-RED + ops add-ons.**